BB84 is the **first and most famous quantum cryptography protocol**, developed by **Charles Bennett** and **Gilles Brassard** in **1984 —** that's why it's called **BB84**.

The BB84 protocol is used to **securely share a secret key** between two people (commonly called Alice and Bob) over an insecure channel using quantum bits (qubits). It ensures that even if someone (Eve) tries to spy, Alice and Bob will **know**.

1. Alice Sends Qubits

- Alice prepares qubits using **random bit values** (0 or 1).
- She **randomly chooses** a basis to encode each bit:
  - **Z basis (+)**: Uses $|0\rangle$ and $|1\rangle$
  - **X basis (×)**: Uses $|+\rangle$ and $|-\rangle$
    - $|+\rangle=(|0\rangle+|1\rangle)/\sqrt{2}$
    - $|-\rangle=(|0\rangle-|1\rangle)/\sqrt{2}$
- She then sends the qubits to Bob over a quantum channel.

2. Bob Measures

- Bob **doesn't know** which basis Alice used.
- So, he **randomly chooses a basis** (Z or X) to measure each incoming qubit.
- If he chooses the **correct basis**, he gets the correct bit.
- If he chooses the **wrong basis**, he gets a **random result** (50% chance of 0 or 1).

3. Compare Bases

- After all qubits are sent, **Alice and Bob talk over a public channel**.
- They **reveal the bases they used, not the bit values**.
- They **keep only the bits where the bases matched**.

4. Error Checking

- Alice and Bob now have a smaller set of **matched bits —** called the **raw key**.
- To check for **eavesdropping**, they:
  - **Publicly reveal some bits** from the raw key.
  - **Compare them** to see if there's a high number of errors.
- If too many errors (e.g., >15%), **they assume Eve was spying** and discard the key.
- If error is low, they **keep the key** and apply **privacy amplification** to make it even more secure.

This table shows how Alice encodes each bit into a quantum state using one of two possible bases — the **Z basis (also called +)** and the **X basis (×)**.

- If Alice chooses the Z basis and the bit is 0, she sends the state $|0\rangle|0\rangle|0\rangle$.
- If the bit is 1, she sends $|1\rangle$.
- But if she uses the X basis instead:
    - A 0 is sent as $|+\rangle$, which is a superposition of $|0\rangle$ and $|1\rangle$.
    - A 1 is sent as $|-\rangle$, which is the opposite superposition.

These states are then sent to Bob, who tries to measure them — but without knowing the basis, he has a 50% chance of guessing wrong. This randomness and the laws of quantum mechanics make BB84 secure.

Slide 10:

Here we see that quantum measurements are **non-deterministic** unless the bases match. Bob uses measurement operators like $P0=|0\rangle\langle 0|$ to project qubits onto classical outcomes.
If Bob uses the wrong basis — like measuring $|+\rangle$ in the Z basis — the probability of getting 0 or 1 is both 50%.
This randomness helps ensure **eavesdroppers** can't gain information without being detected.